LESSON 03 · CHEAT SHEET

Functions

DEFINE, THEN CALL

```
def share(reward, party):     # def name(params):
    """One hero's share, to the penny."""
    return round(reward / party, 2)  # hand back

s = share(1000, 3)               # call: brackets!
print(s)                         333.33
```

VBA: `Function Share(reward, party) … Share = … End Function`. Define **above** the first call.

RETURN VS PRINT

```
def money(x):
    return f"{x:,.2f} cr"    # caller gets the text

def show(x):
    print(f"{x:,.2f} cr")    # caller gets None

money(5) + " net"            '5.00 cr net'
show(5) + " net"             # TypeError: None + str
```

No `return` (or a bare `return`) means the function returns `None`. Return values; print at the end.

EARLY RETURN

```
def purse_rate(quests_done, difficulty):
    if quests_done < 3:
        return 0          # function ends here
    if difficulty >= 4:
        return 0.25
    if difficulty == 3:
        return 0.10
    return 0              # everything else
```

DEFAULTS & KEYWORDS

```
def percent(x, decimals=1):
    return f"{x * 100:.{decimals}f}%"

percent(0.125)               '12.5%'
percent(0.125, 0)            '12%'
percent(0.125, decimals=2)   '12.50%'

def print_notice(quest, reward, party=4, cut=0.1):
    ...
print_notice("Rats", 240, cut=0.05)  # keyword
```

Required params first, defaults after. Keywords can come in any order.

FUNCTIONS CALLING FUNCTIONS

```
def money(x):
    return f"{x:,.2f} cr"

def notice_line(label, amount):
    return f"{label:<15}{money(amount):>12}"

def purse(reward, quests_done, difficulty):
    rate = purse_rate(quests_done, difficulty)
    return reward * rate
```

One rule, one place. Change `money` and every line follows.

SCOPE: INSIDE STAYS INSIDE

```
rate = 0.10                  # outside: seen by all

def guild_cut(reward):
    cut = reward * rate      # can READ rate
    return cut               # local: only in here

def other(reward, rate):     # its own rate wins
    return reward * rate

print(cut)                   # NameError
```

Need something from inside? `return` it. Need something inside? Pass it as a parameter.

TRY FUNCTIONS IN THE REPL

```
python -i my_work\ex5_rations.py
>>> rations_days(6)
28
>>> rations_days(6, base=28)
31
>>> help(rations_days)       # shows the docstring
```

ONE-LINE IF

```
unit = "year" if n == 1 else "years"
```

GOTCHAS

- No `return` → you get `None`
- `money` vs `money(50)`: recipe vs result
- Call before `def` → NameError
- Code after `return` never runs
- Body not indented → IndentationError
- Changing a parameter inside doesn't change the outside variable